

EDUBOT AN AI BASED SMART CHATBOT FOR KNOWLEDGE MANAGEMENT SYSTEM

*A Mini Project report submitted in partial fulfillment of the requirements for the
award of the degree of*

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

by

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VAAGESWARI COLLEGE OF ENGINEERING

UGC AUTONOMOUS

Accredited by NAAC with A+ Grade

(Affiliated to JNTU Hyderabad & Approved by AICTE New Delhi)

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CERTIFICATE

This is to certify that the project report entitled “**EDUBOT AN AI BASED SMART CHATBOT FOR KNOWLEDGE MANAGEMENT SYSTEM**” submitted by the following students in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in CSE(AI&ML) and is a Bonafide record of the work performed by.

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ABSTRACT

The pandemic has brought in a lot of changes in various aspects of human life. Every sector has been affected by this and so is the educational sector. Organizing and accessing of documents was one of the challenges that students faced during the pandemic. To overcome this problem, we propose an AI system that is equipped with a chatbot. In this study, we develop a smart knowledge management system that authenticates the user via face recognition, then a chatbot helps the users in searching and accessing the required document easily. Face Recognition is implemented using libraries such as OpenCV, Tensorflow and Face_Recognition. Face_Recognition Library provided the highest accuracy so it was used in the smart knowledge management system. A chatbot was implemented to search and access the study materials such as notes, textbooks and question banks from a particular destined folder through the hyperlinks created. To make the system user friendly, GUI was designed using Tkinter library. This AI based smart knowledge management system provides access to the required document very effectively and effortlessly.

CONTENTS

TOPICS	Page No's
Certificates	
Acknowledgemnt	
Abstract	
Figures/Tables	
CHAPTER-1: INTRODUCTION	1-3
CHAPTER-2: LITERATURE SURVEY	4-7
CHAPTER-3: SYSTEM ANALYSIS	
3.1 Hardware Requirements	8
3.2 Software Requirements	8-9
3.3 Proposed System	9
3.4 Modules	10
CHAPTER-4: SYSTEM DESIGN	
4.1 System Architecture	11
4.2 Data Flow Diagrams	11-12
4.3 UML Diagrams	12
4.4 Use Case Diagram	13
4.5 Class Diagram	14
4.6 Sequence Diagram	15
4.7 Activity Diagram	15-16
4.8 Collaboration Diagram	17
4.9 ER Diagram	17
4.10 Introduction to Technologies	
4.11 Python	18-25
CHAPTER-5: IMPLEMENTATION	
5.1 Source Code	26
CHAPTER-6: RESULT AND DISCUSSIONS	
6.1 Screenshots	27-37
CHAPTER-7:CONCLUSION	38
7.1 Future Scope	39
REFERENCES	40

LIST OF FIGURES

S.NO	TABLES/FIGURES	PAGE NO'S
1	System Architecture	11
2	UML Diagrams	12
3	Use Case Diagram	13
4	Class Diagram	14
5	Sequence Diagram	15
6	Activity Diagram	15-16
7	Collaboration Diagram	17
8	ER Diagram	17
9	Screenshots	22-28

CHAPTER-1

INTRODUCTION

1.1 ABOUT THE PROJECT

Many things changed during the covid pandemic; one such example is the education system. The delivery of education had to change from offline to online. Theory classes, laboratories, assessments, exams, training and skill development programs, everything shifted to online mode. A lot of ICT tools were explored and utilized. All the knowledge was in the form of documents on the system. The library wasn't available for students and there were no hard-bound books to refer. That's when soft copies of the book and notes became more prominent. Documents were downloaded multiple times from various sources and were saved abruptly. The computer system was flooded with a greater number of notes and books; managing and retrieving the documents effectively and efficiently was a challenge. The searching of the document takes a lot of effort particularly when the documents have been saved multiple times at multiple locations. This results in stress for students during exams. So, to address this issue, there is a need for a smart system that helps to search and retrieve the document effectively and quickly. Therefore, this study proposes a chatbot that is based on Artificial Intelligence (AI) technique to organize, search and retrieve the documents.

To provide secure access to the documents, face recognition is used for authentication. In addition to documents, the proposed AI based chatbot¹ helps the users with cafeteria information (Menu and cost of food items). Currently, chatbots are used by teachers to conduct quizzes to students and their automatic evaluation. But there aren't many chatbots which can interact directly with students or users along with their authentication. Here, a chatbot has been developed using the AI technology that serves the purpose of providing security to the user and easy access of study materials in the form of hyperlink. The paper is organized as below: Section 2 explains the materials and methods; Section 3 describes the results of the chatbot and Section 4 presents the conclusion.

1.2 PROBLEM STATEMENT

In today's fast-paced digital environment, organizations struggle to manage and utilize vast amounts of knowledge effectively. Knowledge management systems (KMS) are crucial for storing and organizing valuable information; however, many existing systems rely on outdated, static methods that fail to provide quick, relevant, and context-specific insights. Employees often waste time searching for information, and knowledge sharing is inefficient due to the limitations of traditional systems.

The lack of personalized, real-time assistance further exacerbates these issues, leading to delays, errors, and low productivity. A key challenge is the absence of an intelligent interface that can provide seamless, natural language-based interactions while retrieving and disseminating knowledge efficiently.

1.2 OBJECTIVE

The objective of this project is to develop an AI-based smart chatbot that enhances the efficiency and user-friendliness of a knowledge management system. The smart chatbot will leverage Natural Language Processing (NLP) and Machine Learning (ML) techniques to:

1. **Automate Knowledge Retrieval:** The chatbot will be capable of answering user queries and retrieving relevant information from the knowledge base in real-time, reducing the need for manual searches.
2. **Personalized User Interaction:** The system will tailor responses based on the user's role, preferences, and previous interactions, making the knowledge retrieval process faster and more relevant.
3. **Continuous Learning:** By integrating machine learning, the chatbot will improve its responses over time by learning from user interactions and adapting to new knowledge inputs.
4. **Improve Knowledge Sharing:** Enable more effective collaboration and knowledge sharing by providing a conversational interface, which will increase user engagement and reduce knowledge silos within the organization.
5. **Enhance Decision-Making:** The chatbot will help users make better-informed decisions by delivering contextually accurate information, actionable insights, and recommendations based on the organization's knowledge base.

CHAPTER-2

LITERATURE SURVEY

Title: ‘A Platform for Human-Chatbot Interaction Using Python’

Abstract: Over recent years, we've seen various customs for conversational agents. Chatbot is a conventional agent which is capable to communicate with operators by using natural languages. As numerous chatbot platforms already exist, there are still some problems in building data-driven system because a huge amount of data is required for its development. Thus, this paper describes various such agents which depend upon natural expressions implemented in Python. Moreover, to provide a better platform, web connectivity is also provided to evaluate the chatbot on a web-based platform which will help in analysing Human-Chatbot interactions.

Title: ‘Deep Learning Techniques for Implementation of Chatbots’

Abstract: Chatbots are software programs that interact with clients using natural languages. The motto of the researchers was to know if chatbots can able to fool the clients that they were real humans. To develop a chatbot that can pass the Turing test, plenty of effort done with the introduction of the ELIZA chatbot in the year 1966. Various approaches for the development of chatbots and different technologies in the creation of chatbots developed because of those efforts. NLTK is a module in python which can able to perform Natural Language Processing. It is used to analyse the input in the form of speech and generate responses that are similar to humans. Nowadays there is a lot of demand for virtual assistants such as Siri, Cortana, Google Assistant and Alexa, and speech-based search engines. Nowadays Chatbots are gaining massive demand mainly in the business sector for automating client service and also for reducing efforts of humans. Chatbots typically used for information acquisition in dialogue systems. To perfectly imitate a human response, a chatbot should examine the query asked by a client correctly and design an appropriate response. In this study we compare and discuss the different technologies used in the chatbots and also address the design and implementation of a chatbot system.

Title: "A Chatbot for Changing Lifestyle in Education,"

Abstract: In this pandemic situation, all are learning education online. There is a lot of drawbacks to these methods, the main drawback of this system is the interaction between students and teachers become low. A chatbot is one of the most convenient ways of studying for students and it also rectifies student doubts at any time without human support. This paper aims to approach a typical way to design a chatbot for MATLAB practical dataset. Students can ask a question in the chatbot in the form of text then, the question is processed with natural language processing and deep learning technology. Finally, the chatbot can answer the students with exact answers. So this kind of chatbot is useful for both the students and teachers.

Title: "FCFD: Teach the machine to accomplish face detection step by step,"

Abstract: In this paper, a novel progressive strategy is proposed to teach the machine to accomplish face detection in the wild. Firstly, deep model named Fully-connected Face Classifier (FCFC) is built up. With the targeted training data, FCFC learns the knowledge corresponding to distinguish face in various pose, facial expression, occlusion proportion, and blur degree from background gradually. Secondly, FCFC is converted to Fully Convolutional Face Detector (FCFD) which is able to handle a face image in arbitrary size. Finally, FCFDs with different receptive fields and sliding strides are combined to detect faces in various sizes for a given image scale. Experiments on FDDB face detection benchmark dataset and social network selfies show that the proposed FCFD achieves promising results, especially in the detection of blurred or occluded face. Generalization of our FCFD to social network selfies is instructive to analyze and retrieve those photos, since more discriminative information of one person (e.g., gender, identity) can be extracted once his or her face is located.

Title: "A Comparative Study on Convolutional Neural Network Based Face Recognition,"

Abstract: This paper presents a comparative study to recognize faces from a customized dataset of 10 identities of different celebrities using Convolutional Neural Network based models such as AlexNet, VGG16, VGG19 and MobileNet. These pre-trained models previously trained on ImageNet dataset are used with the application of Transfer Learning and Fine Tuning. For our experiment we used Keras API with TensorFlow backend written in Python. The performance analysis includes training, validation, and testing on different images created from original dataset. The validation accuracy of VGG19 model is found better than the other three but MobileNet model showed better test accuracy.

Title: "Face Detection and Tracking using OpenCV"

Abstract: An application for tracking and detecting faces in videos and in cameras which can be used for multipurpose activities. The intention of the paper is deep study of face detection using open CV. A tabular comparison is performed in order to understand the algorithms in an easier manner. It talks about various algorithms like Adaboost, Haar cascades. This paper aims to help in understanding the best prerequisites for face detection.

Title: ' Face Detection in Real Time Based on HOG'

Abstract: A new method of face recognition based on gradient direction histogram (HOG) features extraction and fast principal component analysis (PCA) algorithm is proposed to solve the problem of low accuracy of face recognition under non-restrictive conditions. In this method, the Haar feature classifier is used to extract and extract the original data, and then the HOG features are extracted from the image data and the PCA dimension reduction is processed, and the Support Vector Machines (SVM) algorithm is used to recognize the face. The experimental results of the classification recognition on the LFW face database verify the effectiveness of the method.

Title: "Voice Controlled Personal Health Assistance Device using Natural Language Processing and Random Forest Disease Classifier"

Abstract: Health assistant devices are a need of the hour. In this paper, we outline one such device that has been developed. It is voice-controlled device that uses Google's speech to text converter API and Natural Language Processing for the communication with the user. The user provided input symptoms are classified into a disease using a Random Forest classifier. The NLP and classification are performed within the Raspberry Pi which acts as a standalone device. Temperature sensor is also interfaced for better user experience. The developed classifier has the accuracy of 95.02% which has further scope of improvement with a larger data set.

Title: "Deep learning Model Used in Text Classification,"

Abstract: Text classification is one of the most widely used natural language processing technologies. Common text classification applications include spam identification, news text classification, information retrieval, emotion analysis, and intention judgment, etc. Traditional text classifiers based on machine learning methods have defects such as data sparsity, dimension explosion and poor generalization ability, while classifiers based on deep learning network greatly improve these defects, avoid cumbersome feature extraction process, and have strong learning ability and higher prediction accuracy. For example, convolutional neural network (CNN)[1]. This paper introduces the process of text classification and focuses on the deep learning model used in text classification.

Title: "The Open Look graphical user interface and its toolkits,"

Abstract: X11, Open Look and Motif are window and graphical user interface technologies that are now becoming available from vendors. Networked window systems such as X and NeWS allow the user to run their application on one machine and have a multi-window, graphical user interface to that application from their desktop. The new technology of networked window systems is accompanied by second generation graphical user interface designs such as Open Look and Motif. Standard user interface designs allow the user to concentrate on learning about the functionality of a new application, rather than learning an application-specific user interface.

CHAPTER-3

SYSTEM ANALYSIS

3.1 HARDWARE REQUIREMENTS

MINIMUM (Required for Execution)		MY SYSTEM (Development)
System	Pentium IV 2.2 GHz	i3 Processor 5 th Gen
Hard Disk	20 Gb	500 Gb
Ram	1 Gb	4 Gb

3.2 SOFTWARE REQUIREMENTS

Operating System	Windows 10/11
Development Software	Python 3.10
Programming Language	Python
Domain	Machine Learning
Integrated Development Environment (IDE)	Visual Studio Code
Front End Technologies	HTML5, CSS3, Java Script
Back End Technologies or Framework	Django
Database Language	SQL
Database(RDBMS)	MYSQL
Web server or Deployment	Django Application
Server Desing/Modeling	Rational Rose

3.3 PROPOSED SYSTEM

The proposed system, titled "Edu-bot: An AI-based Smart Chatbot for Knowledge Management System," aims to address the challenges faced by students during the pandemic in organizing and accessing educational documents. Leveraging cutting-edge technologies, the system introduces a multifaceted approach. First, user authentication is implemented through face recognition, utilizing OpenCV, TensorFlow, and the Face_Recognition Library to ensure a high level of accuracy. This adds a layer of security and personalization to the knowledge management system. The core functionality of the system involves a chatbot designed to assist users in seamlessly searching and accessing the required study materials, such as notes, textbooks, and question banks. The chatbot relies on hyperlinks created within a designated folder structure, facilitating efficient document retrieval. To enhance user interaction and accessibility, a user-friendly graphical user interface (GUI) is developed using the Tkinter library. The overall goal of the Edu-bot system is to provide students with an effective and effortless means of accessing educational resources, thus mitigating the challenges posed by document organization during the pandemic.

ADVANTAGES

Efficient Document Retrieval:

The system streamlines the process of accessing educational materials by employing a chatbot that uses hyperlinks within a designated folder structure. This efficiency reduces the time and effort required for students to find and retrieve specific study materials.

Enhanced Security with Face Recognition:

The integration of face recognition technology using OpenCV, TensorFlow, and the Face_Recognition Library adds a robust layer of user authentication. This enhances the security of the knowledge management system by ensuring that only authorized users can access the educational resources.

User-Friendly Interface:

The graphical user interface (GUI) designed using the Tkinter library contributes to a user-friendly experience. This feature makes the system accessible to users with varying levels of technical proficiency, promoting ease of navigation and interaction.

Personalized Learning Experience:

The face recognition authentication process allows for a personalized learning experience by ensuring that each user has a unique and secure access point. This personalization can be further extended by tailoring the chatbot's responses to individual user preferences and learning patterns.

Adaptability and Scalability:

The use of advanced technologies, such as TensorFlow and Face_Recognition Library, demonstrates the system's adaptability to state-of-the-art tools. Additionally, the proposed system can be designed to scale effectively, accommodating a growing user base and an expanding repository of educational materials as the need arises.

3.4 Modules

- Face Recognition Module
- Chatbot Module
- Document Management Module
- Graphical User Interface (GUI) Module
- Security and Access Control Module

CHAPTER-4

SYSTEM DESIGN

4.1 System Architecture

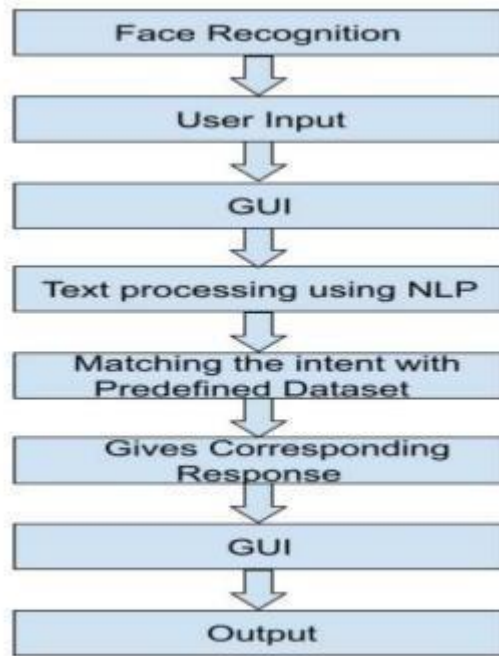


Figure 1: Chatbot workflow

Fig.4.1: System Architecture

4.2 Data Flow Diagrams

1. The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.
2. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
3. DFD shows how the information moves through the system and how it is modified by a series of transformations.

4. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.

4.3 UML Diagrams

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns

4.4 USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

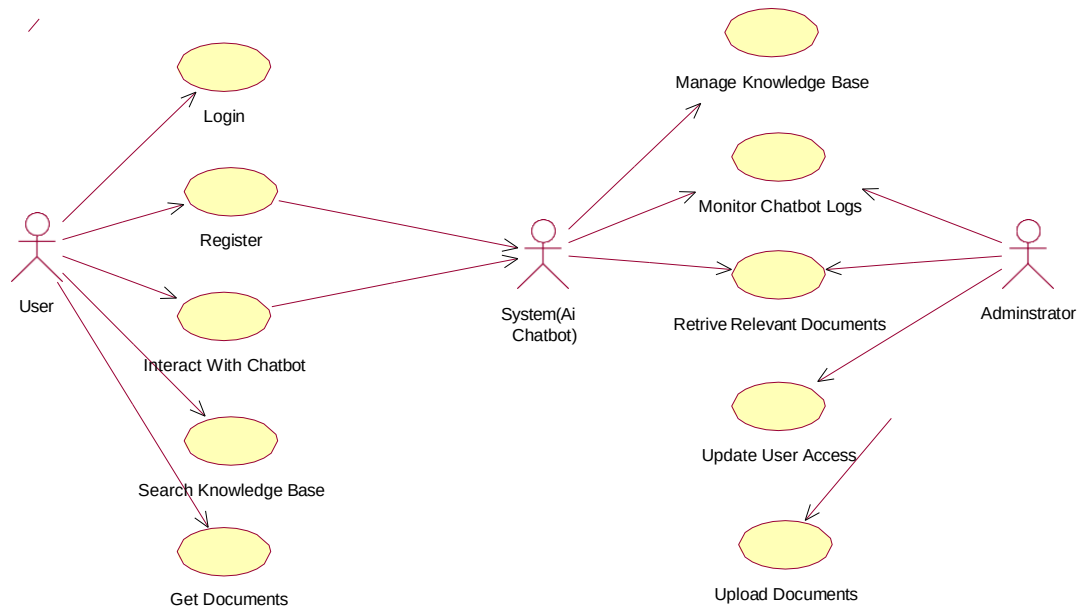


Fig.4.4: Use Case Diagram

4.5 CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

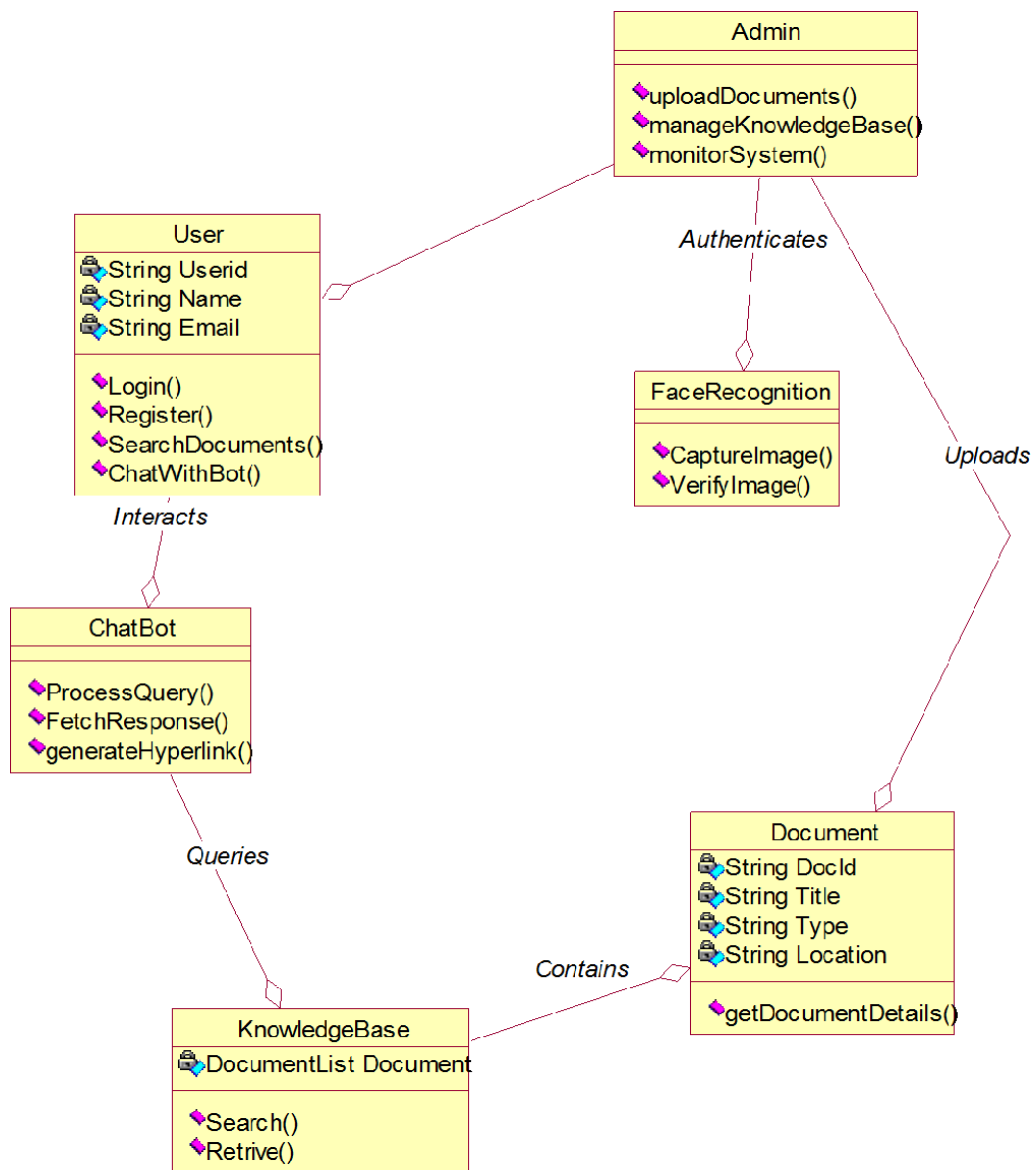


Fig 4.5: Class Diagram

4.6 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

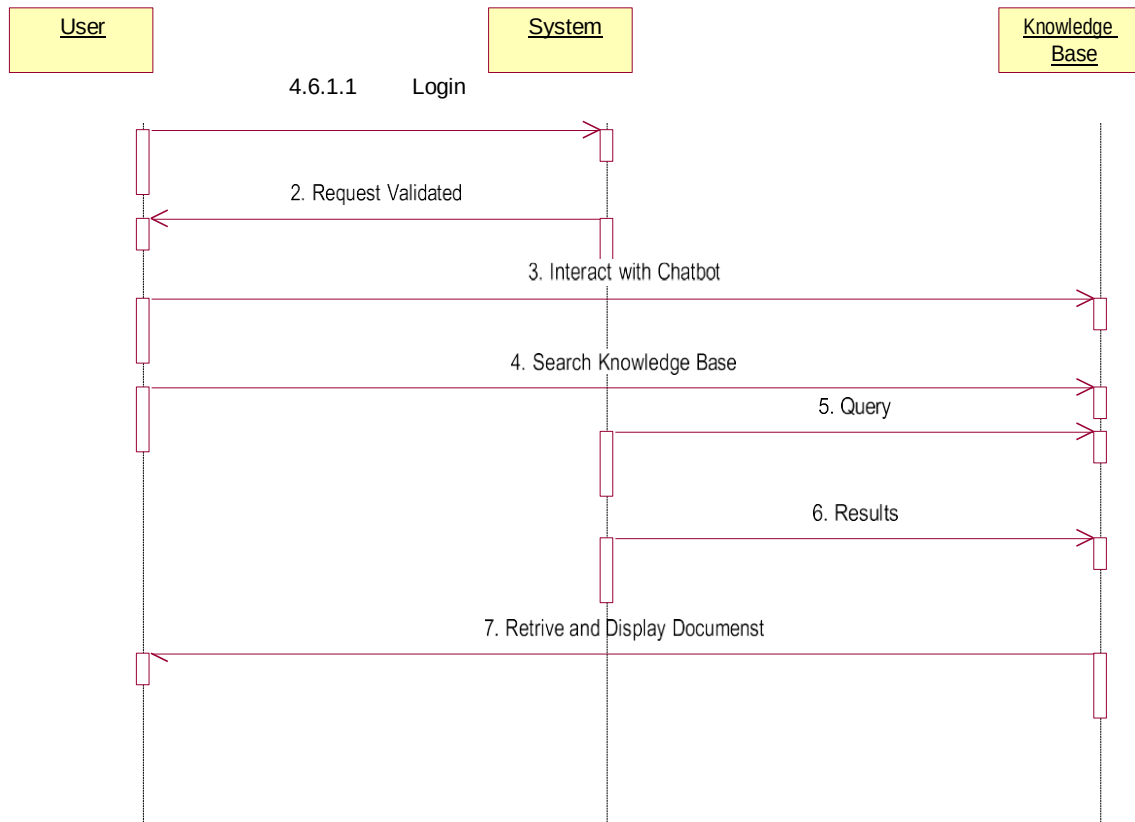


Fig 4.6 :Sequence Diagram

4.7 ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

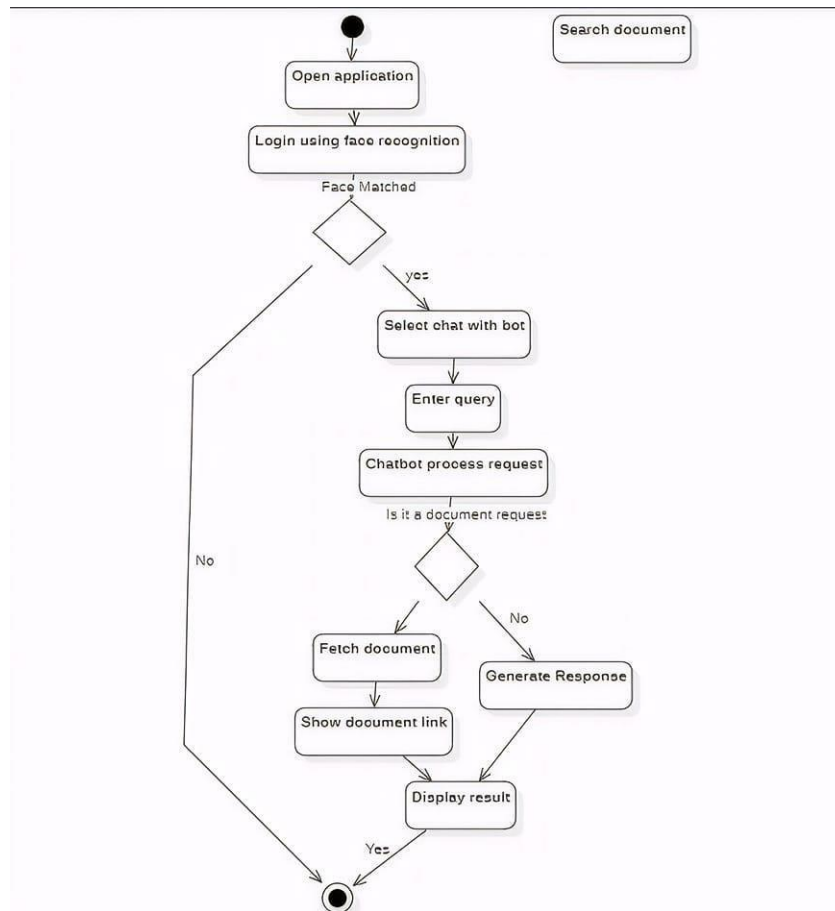


Fig 4.7: Activity Diagram

4.8 COLLABORATION DIAGRAM

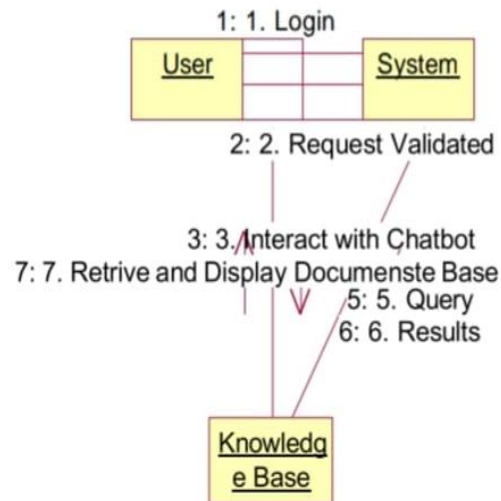


Fig 4.8: Collaboration Diagram

4.9ER-DIAGRAM

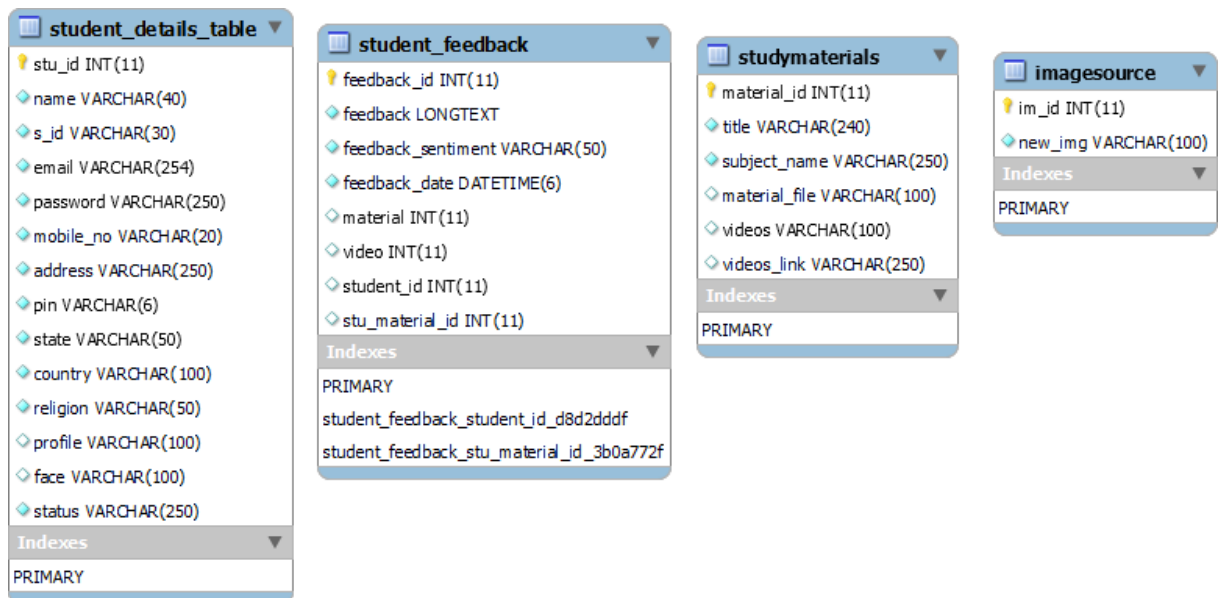


Fig 4.9 : ER Diagram

4.10 INTRODUCTION TO TECHNOLOGIES

4.11 PYTHON

4.11.1 Below are some facts about Python.

4.11.2 Python is currently the most widely used multi-purpose, high-level programming language.

4.11.3 Python allows programming in Object-Oriented and Procedural Paradigm.

Python programs generally are smaller than other programming languages like Java.

4.11.4 Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.

4.11.5 Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber... etc.

The biggest strength of Python is huge collection of standard library which can be used for the following –

4.11.5.1 Machine Learning

4.11.5.2 GUI Applications (like Kivy, Tkinter, PyQt etc.)

4.11.5.3 Web frameworks like Django (used by YouTube, Instagram)

4.11.5.4 Image processing (like Opencv, Pillow)

4.11.5.5 Web scraping (like Scrapy, BeautifulSoup, Selenium)

4.11.5.6 Test frameworks

4.11.5.7 Multimedia

ADVANTAGES OF PYTHON

Advantages of Python :-

Let's see how Python dominates over other languages.

Extensive libraries

Python downloads with an extensive library and it contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more. So, we don't have to write the complete code for that manually.

Extensible

As we have seen earlier, Python can be **extended to other languages**. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add **scripting capabilities** to our code in the other language.

Improved productivity

The language's need to be in simplicity and extensive libraries render programmers **more productive** than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

IoT opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world.

Simple and easy

When working with Java, you may have to create a class to print '**Hello World**'. But in Python, just a print statement will do. It is also quite **easy to learn, understand, and code**. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and **indentation is mandatory**. This further aids the readability of the code.

Object-oriented

This language supports both the **procedural and object-oriented** programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the **encapsulation of data** and functions into one.

Free and open-source

Like we said earlier, Python is **freely available**. But not only can you **download Python** for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to **code only once**, and you can run it anywhere. This is called **Write Once Run Anywhere (WORA)**. However, you need to be careful enough not to include any system dependent features.

Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, **debugging is easier** than in compiled languages.

Advantages of Python Over Other Languages

1. Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in

other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform data analysis and **machine learning**, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in **slow execution**. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the **client-side**. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called **Carbonnelle**.

The reason it is not so famous despite the existence of Brython is that it isn't that secure.

3. Design Restrictions

As you know, Python is **dynamically-typed**. This means that you don't need to declare the type of variable while writing the code. It uses **duck-typing**. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can **raise run-time errors**.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like **JDBC(Java DataBase Connectivity)** and **ODBC (Open DataBase Connectivity)**, Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

This was all about the Advantages and Disadvantages of Python Programming Language.

HISTORY OF PYTHON

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general-purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde & Informatica). The greatest achievement of ABC was to

influence the design of Python. Python was conceptualized in the late 1980s. Guido van Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating system. In an interview with Bill Venners¹, Guido van Rossum said: "In the early 1980s, I worked as an implementer on a team building a language called ABC at Centrum Wiskunde en Informatica (CWI). I don't know how well people know ABC's influence on Python. I try to mention ABC's influence because I'm indebted to everything I learned during that project and to the people who worked on it. Later on in the same Interview, Guido van Rossum continued: "I remembered all my experience and some of my frustration with ABC. I decided to try to design a simple scripting language that possessed some of ABC's better properties, but without its problems. So I started typing. I created a simple virtual machine, a simple parser.

4.12 MACHINE LEARNING

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't. Machine learning is often categorized as a subfield of artificial intelligence, but I find that categorization can often be misleading at first brush. The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of building models of data.

Fundamentally, machine learning involves building mathematical models to help understand data. "Learning" enters the fray when we give these models *tunable parameters* that can be adapted to observed data; in this way the program can be considered to be "learning" from the data. Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data. I'll leave to the reader the more philosophical digression regarding the extent to which this type of mathematical, model-based "learning" is similar to the "learning" exhibited by the human brain. Understanding the problem setting in machine learning is essential to using these tools effectively, and so we will start with some broad categorizations of the types of approaches we'll discuss here.

Categories Of Machine Learning

At the most fundamental level, machine learning can be categorized into two main types: supervised learning and unsupervised learning.

Supervised learning involves somehow modeling the relationship between measured features of data and some label associated with the data; once this model is determined, it can be used to apply labels to new, unknown data. This is further subdivided into classification tasks and regression tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities. We will see examples of both types of supervised learning in the following section.

Unsupervised learning involves modeling the features of a dataset without reference to any label, and is often described as "letting the dataset speak for itself." These models include tasks such as clustering and dimensionality reduction. Clustering algorithms identify distinct groups of data, while dimensionality reduction algorithms search for more succinct representations of the data. We will see examples of both types of unsupervised learning in the following section.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and haven't surpassed human intelligence in many aspects. Then the question is that what is the need to make machine learn? The most suitable reason for doing this is, "to make decisions, based on data, with efficiency and scale".

Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programming logic, in the problems that cannot be programmed inherently. The fact is that we can't do without human intelligence, but other aspect is that

we all need to solve real- world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning

While Machine Learning is rapidly evolving, making significant strides with cybersecurity and autonomous cars, this segment of AI as whole still has a long way to go. The reason behind is that ML has not been able to overcome number of challenges. The challenges that ML is facing currently are -

Quality of data – Having good-quality data for ML algorithms is one of the biggest challenges. Use of low-quality data leads to the problems related to data preprocessing and feature extraction.

Time-Consuming task – Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and retrieval.

Lack of specialist persons – As ML technology is still in its infancy stage, availability of expert resources is a tough job.

No clear objective for formulating business problems – Having no clear objective and well -defined goal for business problems is another key challenge for ML because this technology is not that mature yet.

Issue of overfitting & underfitting – If the model is overfitting or underfitting, it cannot be represented well for the problem.

Curse of dimensionality – Another challenge ML model faces is too many features of data points. This can be a real hindrance.

Difficulty in deployment – Complexity of the ML model makes it quite difficult to be deployed in real life.

Applications of Machine Learning :-

Machine Learning is the most rapidly growing technology and according to researchers we are in the golden year of AI and ML. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML

- Emotion analysis
- Sentiment analysis
- Error detection and prevention
- Weather forecasting and prediction
- Stock market analysis and forecasting
- Speech synthesis
- Speech recognition
- Customer segmentation
- Object recognition
- Fraud detection
- Fraud prevention
- Recommendation of products to customer in online shopping

How to Start Learning Machine Learning?

Arthur Samuel coined the term “**Machine Learning**” in 1959 and defined it as a “**Field of study that gives computers the capability to learn without being explicitly programmed**”.

And that was the beginning of Machine Learning! In modern times, Machine Learning is one

of the most popular (if not the most!) career choices. According to Indeed, Machine Learning Engineer Is The Best Job of 2019 with a 344% growth and an average base salary of **\$146,085** per year.

But there is still a lot of doubt about what exactly is Machine Learning and how to start learning it? So this article deals with the Basics of Machine Learning and also the path you

can follow to eventually become a full-fledged Machine Learning Engineer. Now let's get started!!!

How to start learning ML?

This is a rough roadmap you can follow on your way to becoming an insanely talented Machine Learning Engineer. Of course, you can always modify the steps according to your needs to reach your desired end-goal!

Step 1 – Understand the Prerequisites

In the case, you are a genius, you could start ML directly but normally, there are some prerequisites that you need to know which include Linear Algebra, Multivariate Calculus, Statistics, and Python. And if you don't know these, never fear! You don't need Ph.D. degree in these topics to get started but you do need a basic understanding.

(a) Learn Linear Algebra and Multivariate Calculus

Both Linear Algebra and Multivariate Calculus are important in Machine Learning. However, the extent to which you need them depends on your role as a data scientist. If you

are more focused on application heavy machine learning, then you will not be that heavily focused on maths as there are many common libraries available. But if you want to focus on R&D in Machine Learning, then mastery of Linear Algebra and

(b) Learn Statistics

Data plays a huge role in Machine Learning. In fact, around 80% of your time as an ML expert will be spent collecting and cleaning data. And statistics is a field that handles the collection, analysis, and presentation of data. So it is no surprise that you need to learn it!!! Some of the key concepts in statistics that are important are Statistical Significance, Probability Distributions, Hypothesis Testing, Regression, etc. Also, Bayesian Thinking is also a very important part of ML which deals with various concepts like Conditional Probability, Priors, and Posteriors, Maximum Likelihood, etc.

(c) Learn Python

Some people prefer to skip Linear Algebra, Multivariate Calculus and Statistics and learn them as they go along with trial and error. But the one thing that you absolutely cannot skip is Python! While there are other languages you can use for Machine Learning like R, Scala, etc. Python is currently the most popular language for ML. In fact, there are many Python libraries that are specifically useful for Artificial Intelligence and Machine Learning such as Keras, TensorFlow, Scikit-learn, etc. So if you want to learn ML, it's best if you learn Python! You can do that using various online resources and courses such as **Fork Python** available Free on GeeksforGeeks

Step 2 – Learn Various ML Concepts

Now that you are done with the prerequisites, you can move on to actually learning ML (Which is the fun part!!!) It's best to start with the basics and then move on to more complicated stuff. Some of the basic concepts in ML are:

(a) Terminologies of Machine Learning

- **Model** – A model is a specific representation learned from data by applying some machine learning algorithm. A model is also called a hypothesis.
- **Feature** – A feature is an individual measurable property of the data. A set of numeric features can be conveniently described by a feature vector. Feature vectors are fed as input to the model. For example, in order to predict a fruit, there may be features like color, smell, taste, etc.
- **Target (Label)** – A target variable or label is the value to be predicted by our model. For the fruit example discussed in the feature section, the label with each set of input would be the name of the fruit like apple, orange, banana, etc.
- **Training** – The idea is to give a set of inputs(features) and it's expected outputs(labels), so after training, we will have a model (hypothesis) that will then map new data to one of the categories trained on.

- **Prediction** – Once our model is ready, it can be fed a set of inputs to which it will provide a predicted output(label).

(b)Types of Machine Learning

- **Supervised Learning** – This involves learning from a training dataset with labeled data using classification and regression models. This learning process continues until the required level of performance is achieved.
- **Unsupervised Learning** – This involves using unlabelled data and then finding the underlying structure in the data in order to learn more and more about the data itself using factor and cluster analysis models.
- **Semi-supervised Learning** – This involves using unlabelled data like Unsupervised Learning with a small amount of labeled data. Using labeled data vastly increases the learning accuracy and is also more cost-effective than Supervised Learning.
- **Reinforcement Learning** – This involves learning optimal actions through trial and error. So the next action is decided by learning behaviors that are based on the current state and that will maximize the reward in the future.

ADVANTAGES & DISADVANTAGES OF ML

Advantages of Machine learning :-

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation)

With ML, you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus softwares. they learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement

As **ML algorithms** gain experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say you need to make a weather forecast model. As the amount of data you have keeps growing, your algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data

Machine Learning algorithms are good at handling data that are multi-dimensional and multi variety, and they can do this in dynamic or uncertain environments.

5. Wide Applications

You could be an e-tailer or a healthcare provider and make ML work for you. Where it does apply, it holds the capability to help deliver a much more personal experience to customers while also targeting the right customers.

Disadvantages of Machine Learning :-

1. Data Acquisition

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for you.

3. Interpretation of Results

Another major challenge is the ability to accurately interpret results generated by the algorithms. You must also carefully choose the algorithms for your purpose.

4. High error-susceptibility

Machine Learning is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time.

CHAPTER-5

IMPLEMENTATION

5.1 Sample Code

```
#!/usr/bin/env python
"""Django's command-line utility for administrative tasks."""
import os
import sys

def main():
    """Run administrative tasks."""
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'SmartChatbox.settings')
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
            "forget to activate a virtual environment?"
        ) from exc
    execute_from_command_line(sys.argv)

if __name__ == '__main__':
    main()
```

CHAPTER-6

OUTPUT SCREENS

6.1 Screenshots

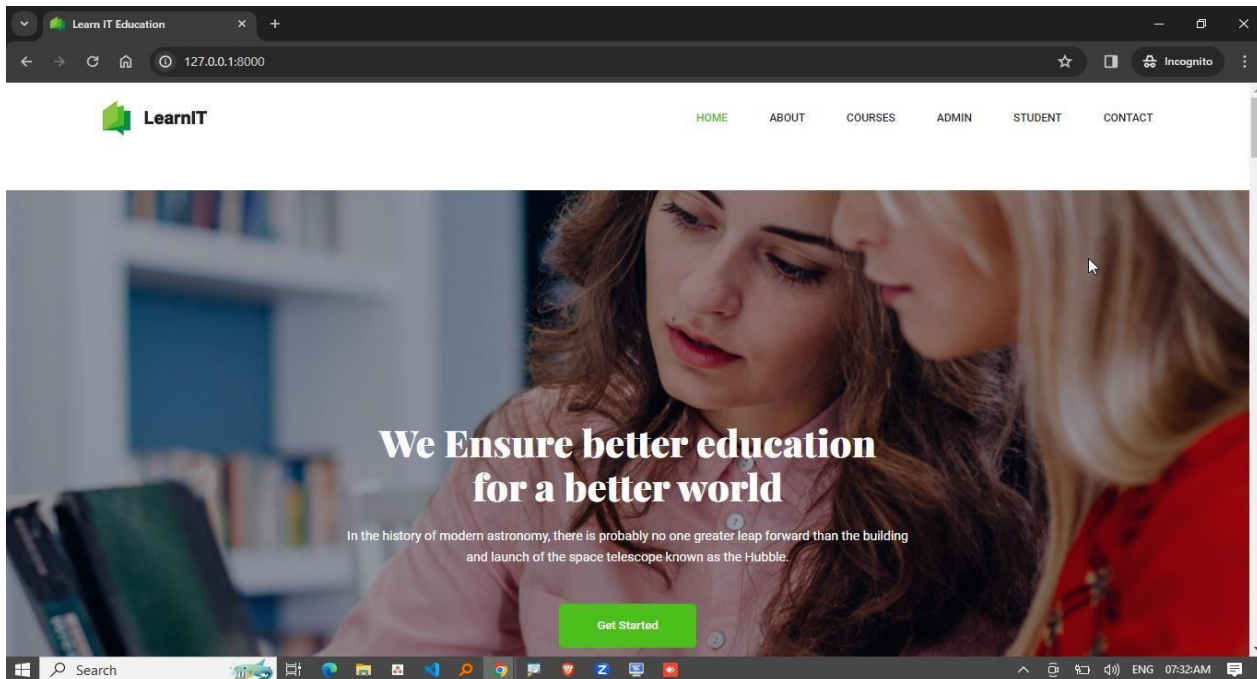


Fig 1:Landing Page

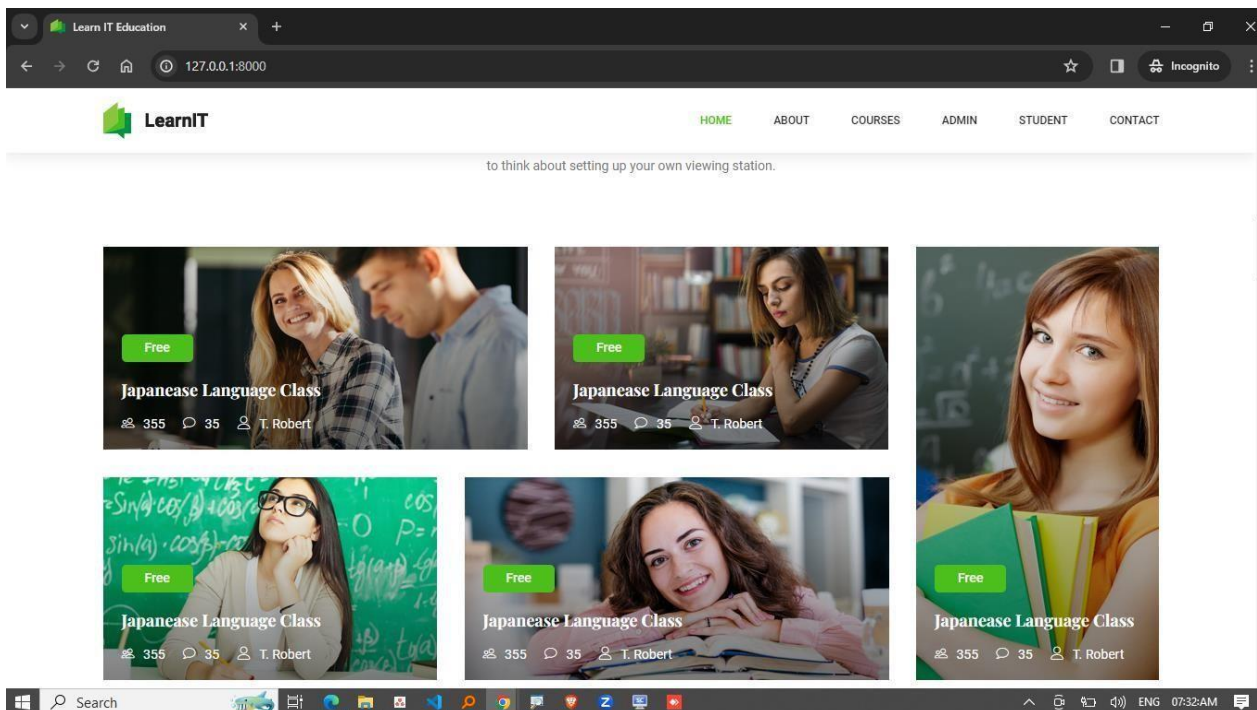


Fig 2: Home Page

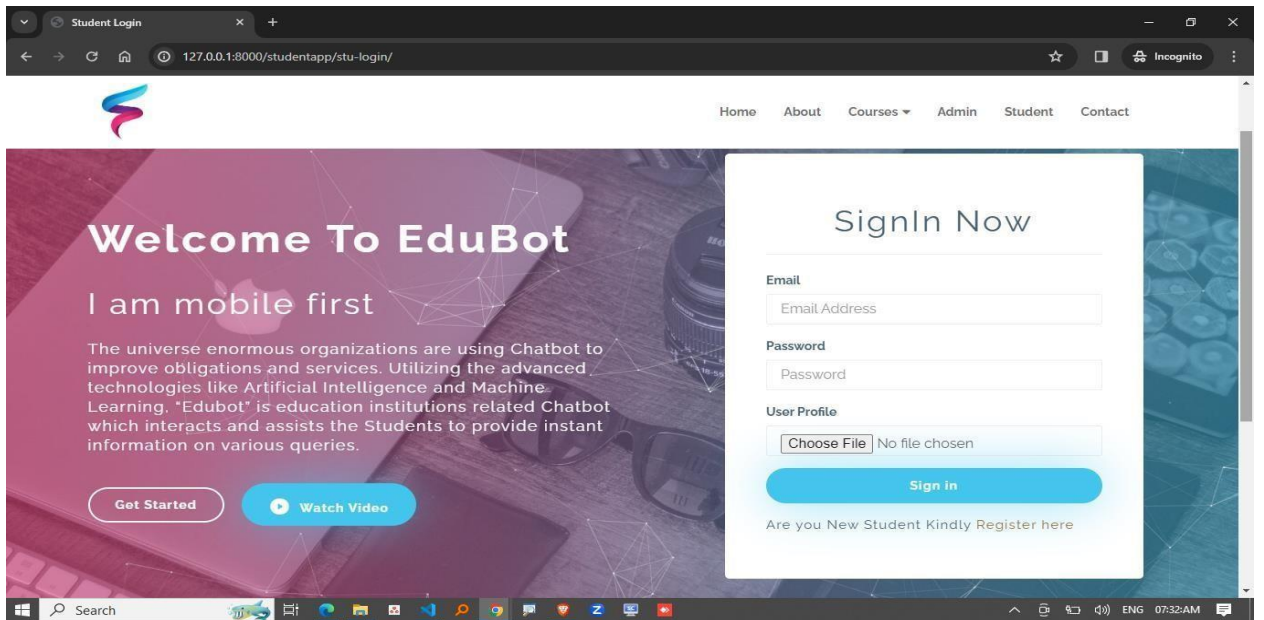


Fig 3: Login Screen

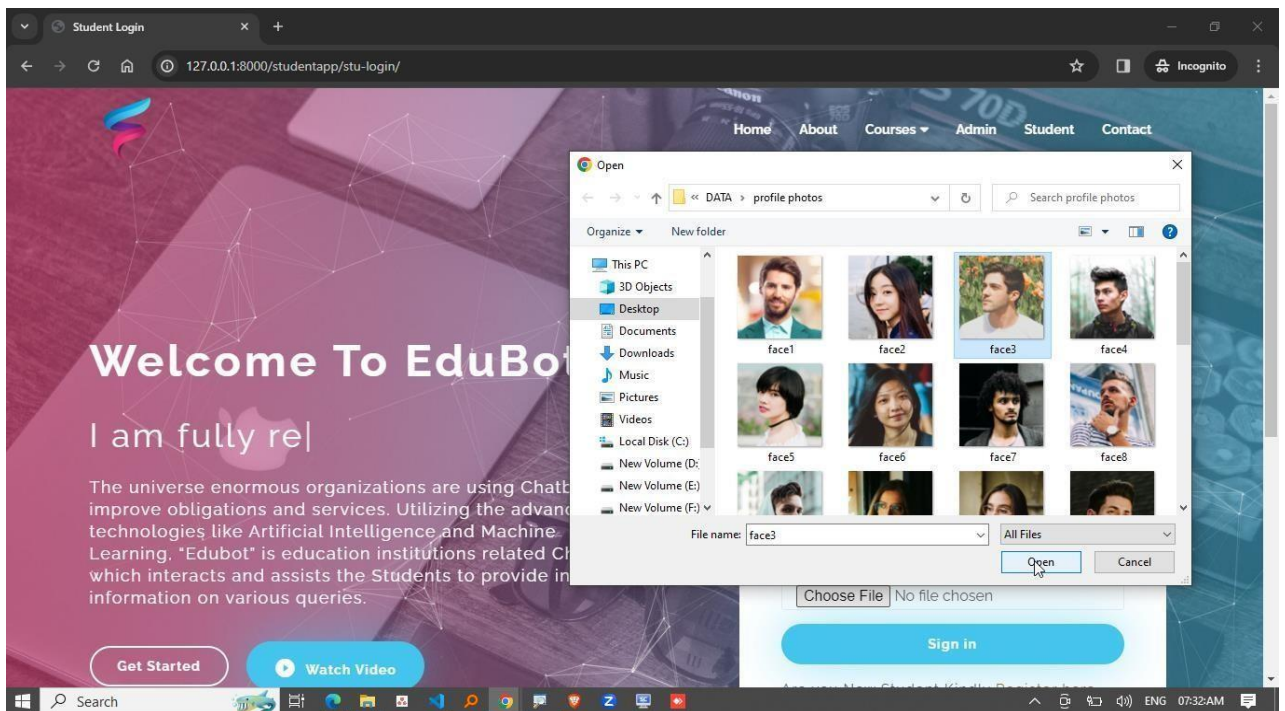


Fig 4: User Profile Upload

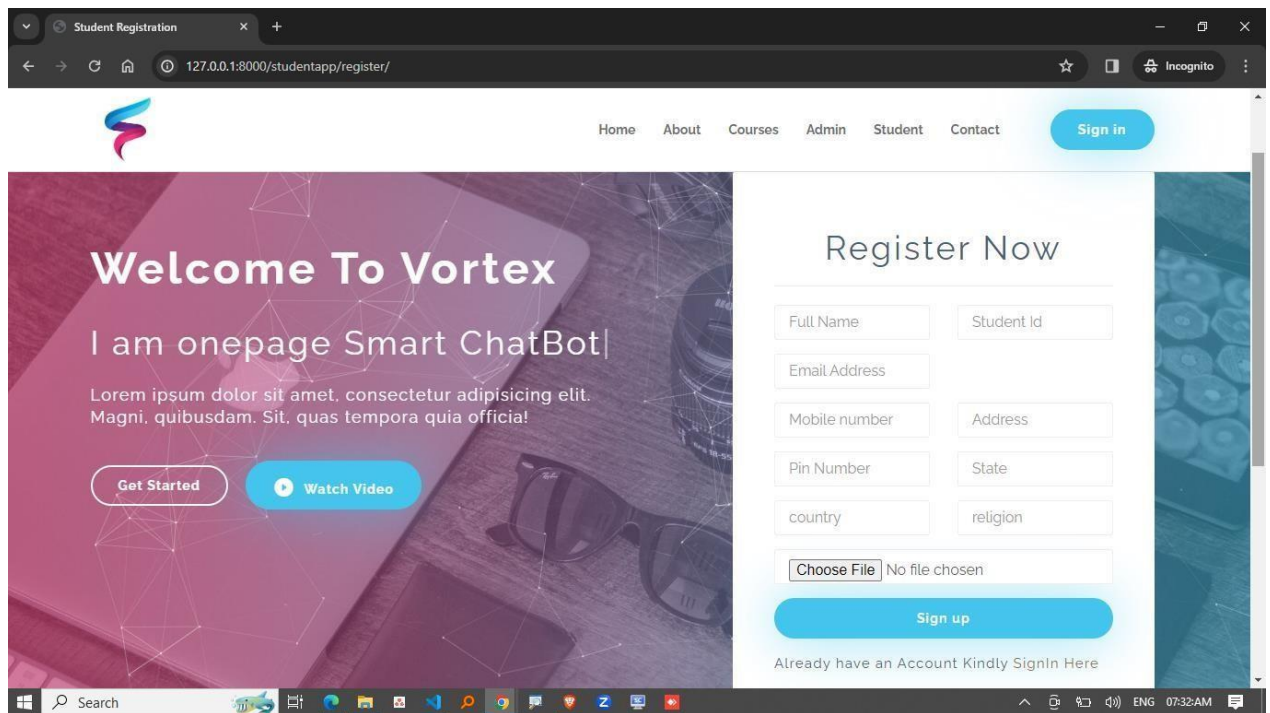


Fig 5: Registration Form Screen

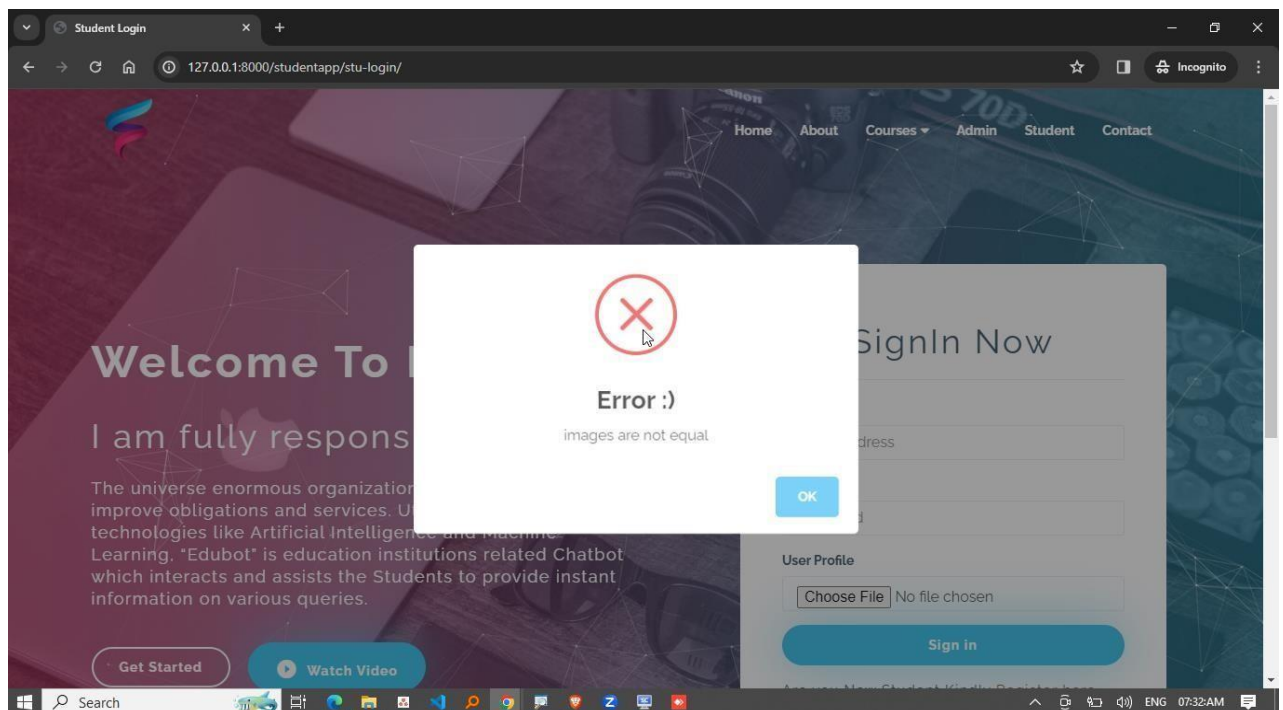


Fig 6: Error Screen

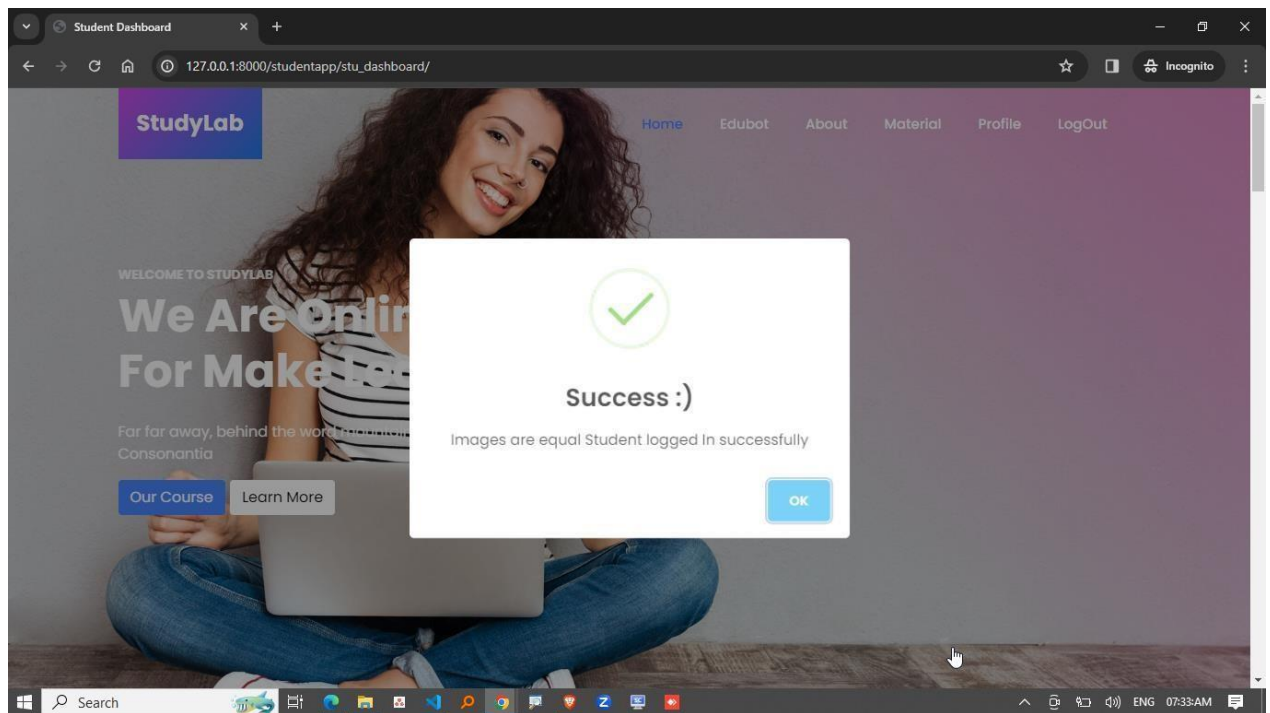


Fig 7: Registration Success Screen

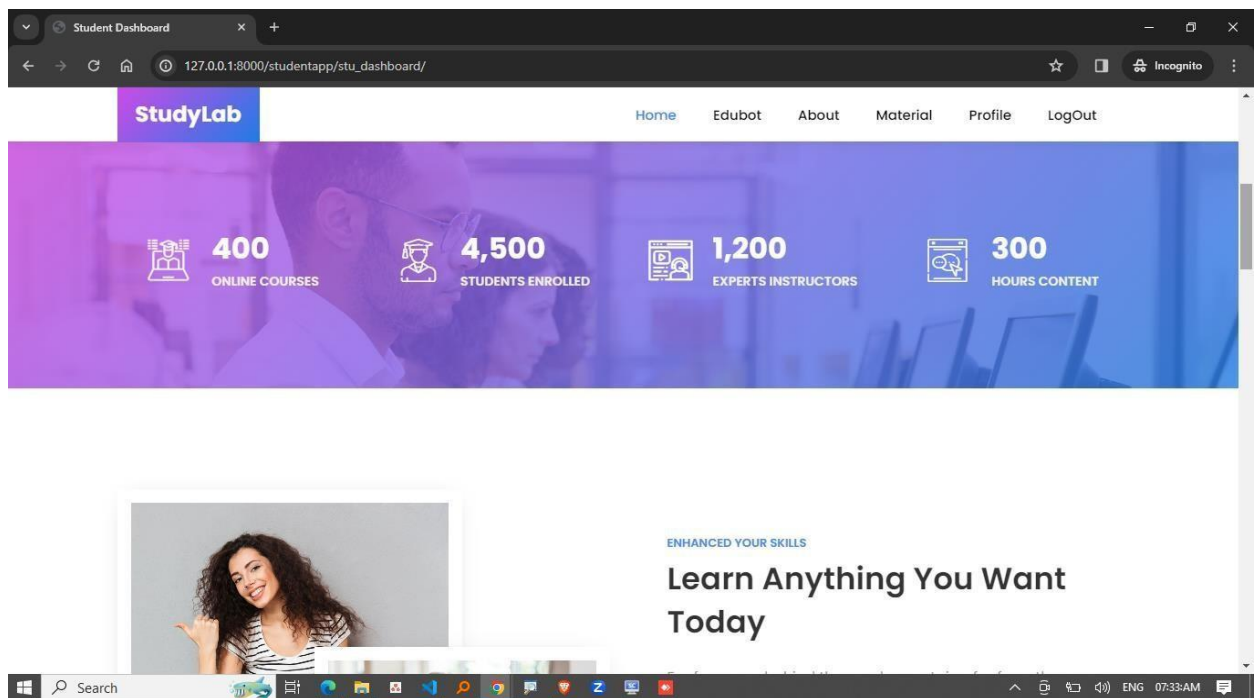


Fig 8: Home Screen

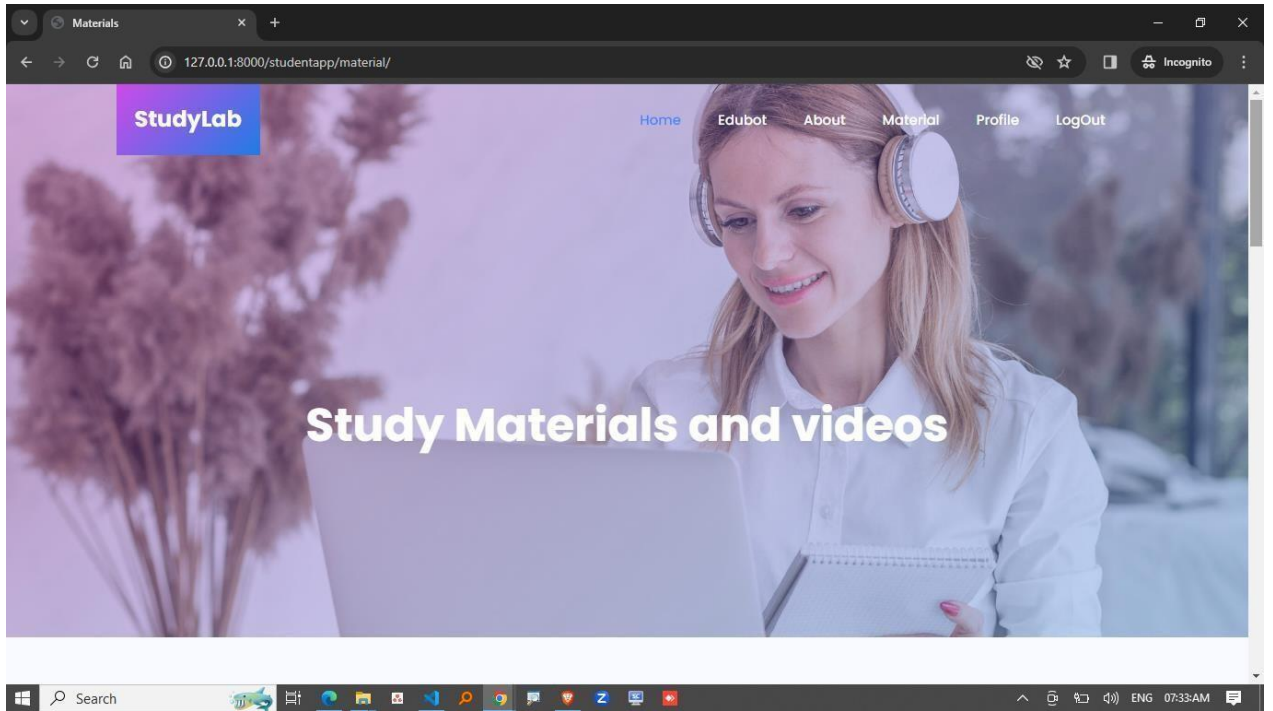


Fig 9: Edubot Screen

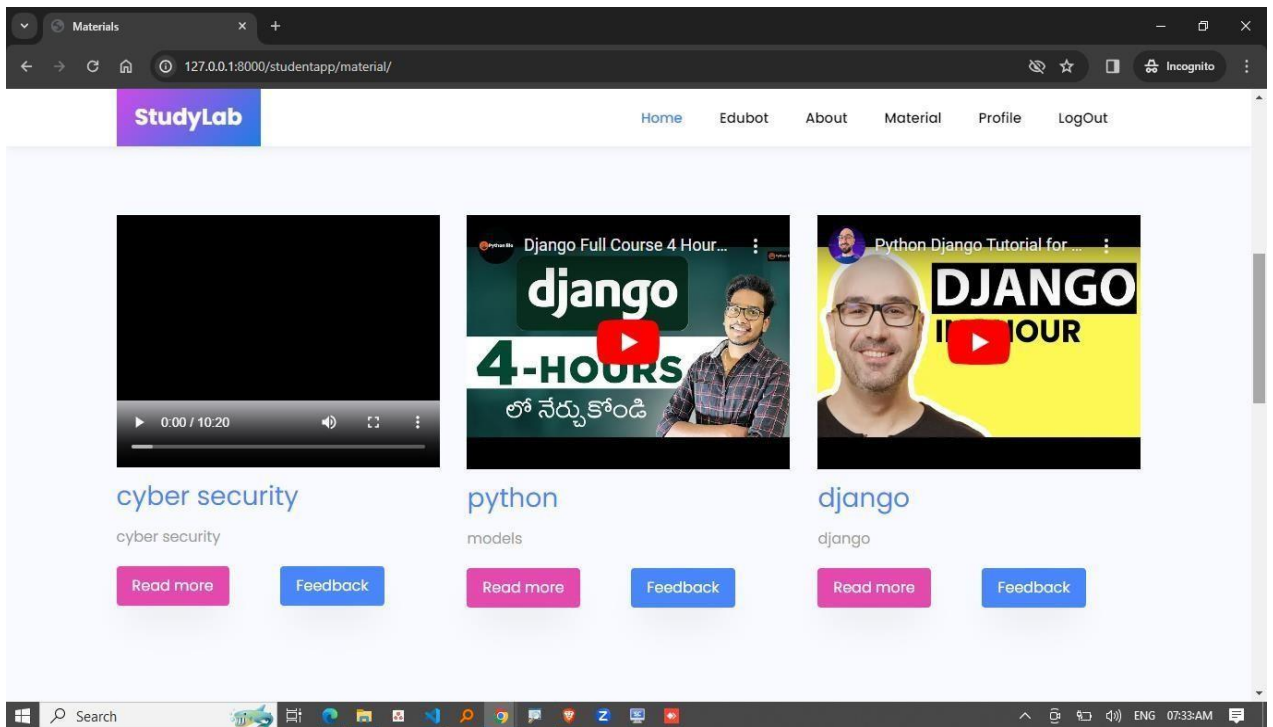


Fig 10: Online Classes Screen

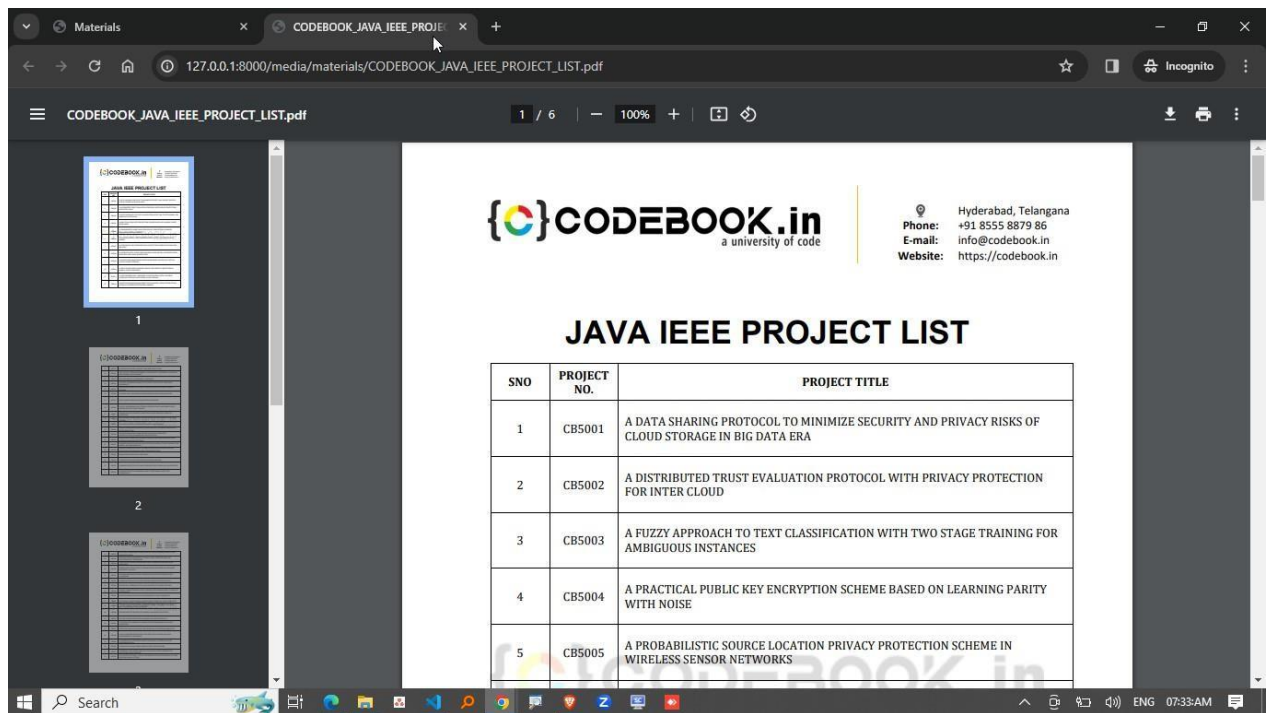


Fig 11: Notes

StudyLab Home Edubot About Material Profile LogOut

1. How do you rate for Study material

★ ★ ★ ★ ★

2. How do you rate for online videos

★ ★ ★ ★ ★

Leave a message here

Fig 12: Feedback Form

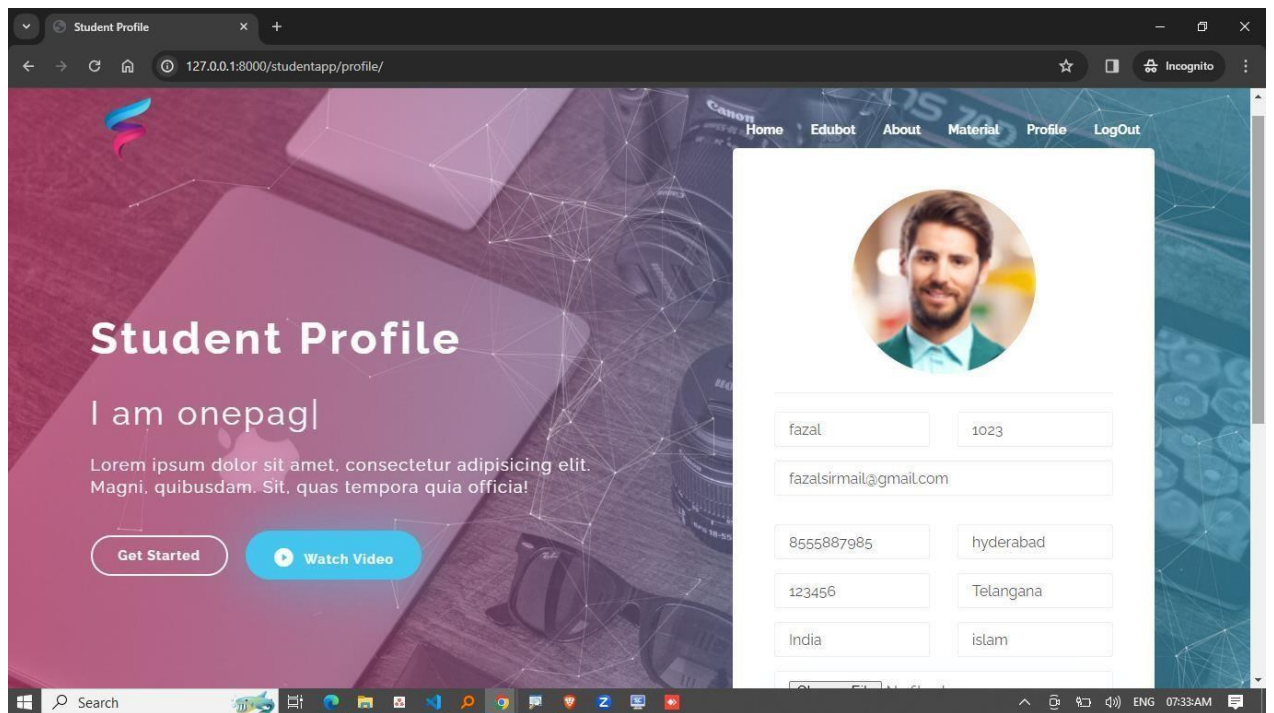


Fig 13: Profile Screen

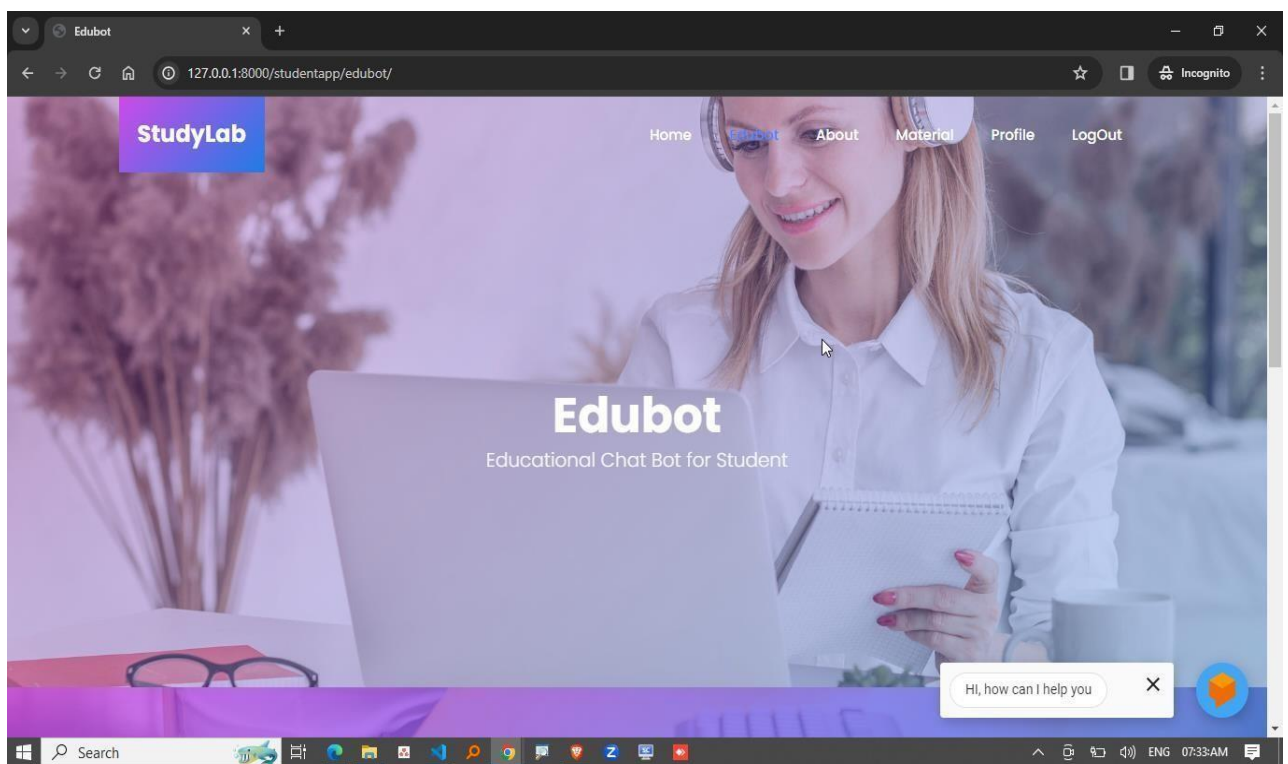


Fig 14: Chatbot

CHAPTER-7

CONCLUSION

Currently, the chatbots are used in most of the industries such as business, e-commerce, etc. The AI based smart chatbot system is designed to be used in educational institutions by students to access their study materials and videos. This will help students to get all the required materials during the examination which will help them to prepare well. Additionally, students' details will also be displayed.

The usage of this knowledge management system results in saving a lot of time and money as the chatbot eliminates the effort to search for examination materials manually. This avoids the necessity to take printouts of the documents. All the materials and details will be organized in a specific manner which is easy to handle.

One of the main advantages of this bot is that it can be used in any domain by just changing the dataset. It can be implemented for different organizations by uploading corresponding datasets. Also, it can be either developed as an app if required or can be implemented in any of the organization's websites. Changing the training algorithm and increasing the number of training steps will improve the accuracy. This chatbot provides cost effective solution since there isn't much hardware involved.

The bot is authenticated through face recognition. When the bot is unlocked, the user is given two options. He can either get access to multiple documents or access documents individually. Multiple documents can be obtained by just clicking on the hyperlink in the GUI else the user can access the individual documents by interacting with the bot. If another user tries to access the bot, the kernel will die and hence security of the chatbot is maintained.

In future study, this bot can be extended to give more precise and effective responses by giving speech as an input. It can be integrated in institution's websites further helping students to use it easily.

7.1 Future Scope

The future scope of an AI-based smart chatbot for a knowledge management system is vast and promising. One significant enhancement could be the incorporation of multilingual support, enabling the chatbot to serve global teams and individuals from diverse linguistic backgrounds. Additionally, the integration with other enterprise tools, such as CRM systems and project management software, could create a seamless user experience by providing access to knowledge across multiple platforms. Advancing the chatbot's natural language understanding (NLU) to handle more complex and nuanced queries, including slang or informal language, would further enhance its ability to respond accurately and contextually. Another promising feature is voice-based interaction, which would make the system more accessible, particularly for users who prefer speaking over typing. Predictive analytics and recommendation features could allow the chatbot to proactively suggest relevant knowledge before users even ask, boosting efficiency and decision-making. Furthermore, incorporating sentiment analysis would enable the chatbot to gauge emotional tone and adjust its responses, offering empathetic support when needed. Integrating with AI-powered document understanding technologies could allow the chatbot to handle unstructured content such as scanned documents and images, improving the richness of its responses. Future versions might also support automated content creation, keeping knowledge bases up-to-date by summarizing complex documents and reports. Collaboration and social features could enhance knowledge sharing, fostering real-time discussions and interactions. Security enhancements, such as role-based access control and data encryption, would be crucial for protecting sensitive information. Proactively delivering relevant updates and knowledge, based on the latest documents or policies, could further streamline workflows. Finally, integrating the chatbot with advanced AI-driven decision support systems would enable it to assist with data-driven decision-making, providing insights and forecasts. These advancements would not only improve knowledge management but also empower organizations with a smarter, more efficient tool for decision-making and collaboration.

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